Exponential_Families

Math 742

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What is the big picture?

We want to be able to put many distributions into a common form, it will make things easier later. Every distribution we have covered so far are part of a family, called the exponential family. Essentially, we want to write the PMF or PDF in a way that separates the data part from the parameter part.

There are 2 ways to do this. Wasserman presents one way (Canonical/natural parameter form) and Casella/Berger way (factorization form).

Definition 1: Canonical/natural parameter form.

A family of PMFs or PDFs are called an **exponential family** if it can be expressed as:

$$p(x|\eta) = h(x)\exp(\eta T(X) - A(\eta)) = h(x)\exp(\eta T(x)\exp(-A(\eta))$$

The Wasserman form has 3 Factors:

1. The base measure $h(x)$. The base measure depends only on the data and not the parameters. It represents the "shape" of the distribution before the parameter shifts it.
2. The exponential tilting term $\exp(\eta T(x))$. η is the natural parameter, it depends on the model parameter. $T(x)$ which is a sufficient statistic (more on sufficiency later). This term shifts and tilts the distribution. For a more detailed explanation see appendix.

$$\rightarrow \text{Tilting} = \text{exponential reweighting}$$

3. The normalizing term $\exp(-A(\eta))$. Ensures the density sums up to 1. It depends only on the the parameter. It is equivalent to the $c(\theta)$ in the Factorization form (i.e. Casella/Berger form).

$$p(x|\eta) = \underbrace{h(x)}_{\text{depends only on } x} \cdot \underbrace{\exp(\eta T(x))}_{\text{tilting}} \cdot \underbrace{\exp(-A(\eta))}_{\text{normalizer}}.$$

Let's immediately apply the definition to a concrete example.

Poisson PMF: $\mathbb{P}(X = x|\lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$

Using the Canonical/Natural parameter form, the Poisson PMF is part of the exponential family because it can be written in this form. We just have to figure out the parts.

$$p(x|\eta) = h(x)\exp(\eta T(X) - A(\eta))$$

First, rewrite λ^x using a log-parameter. Set $\eta = \log(\lambda)$. Then, $\lambda = e^\eta \rightarrow \lambda^x = (e^\eta)^x = e^{\eta x}$.

Using this, the PMF is now,

$$\mathbb{P}(X = x|\lambda) = \frac{1}{x!}\exp(\eta x - e^{\eta})$$

This should make sense, remember $\eta = \log(\lambda)$, so $e^{\eta x} = e^{x\log(\lambda)} = \lambda^x$. So, $\exp(\eta x - e^{\eta}) = e^{x\log(\lambda)}e^{-\lambda} = \lambda^x e^{-\lambda}$.

Now, write down the parts of the Wasserman form,

$$h(x) = \frac{1}{x!}$$

$$T(x) = x$$

Notice that the exponent on the first exponential is ηx, so this is really $\exp(\eta T(x))$.

The remaining term is $\exp(-A(\eta)) = \exp(-e^{\eta})$, so

$$A(\eta) = e^{\eta}$$

This gives us,

$$\mathbb{P}(X = x|\eta) = h(x)\exp(\eta T(x) - A(\eta))$$

$$= h(x)\exp(\eta x - A(\eta))$$

Definition 2: Factorization Form

In the Casella/Berger form, a family of PMFs or PDFs are called an **exponential family** if it can be expressed as:

$$f(x|\theta) = h(x)c(\theta)\exp\left(\sum_{i=1}^{k} w_i(\theta)t_i(x)\right)$$

Note: Usually, in most distributions that we use (e.g. Binomial, Poisson) $k = 1$.

where, $h(x) \geq 0$, $t_1(x), t_2(x), ..., t_k(x)$ are real-valued functions of x that do not depend on θ. $c(\theta) \geq 0$ and $w_1(\theta), w_2(\theta), ..., w_k(\theta)$ are real-valued functions of the possibly vector valued parameter (θ). The θs cannot depend on x.

To show that a family of PDFs or PMfs is an exponential family, we need to identify the functions $h(x), c(\theta), w_i(\theta)$ and $t_i(x)$.

For the Poisson distribution the PMF is,

$$p(x|\lambda) = \frac{1}{x!}e^{-\lambda}\lambda^x$$

The data part is $h(x) = \frac{1}{x!}$.

$$c(\theta) = c(\lambda) = e^{-\lambda}, \theta = \lambda$$
$$\lambda^x = e^{x\log\lambda}$$

$$w(\lambda) = \log(\lambda), t(x) = x$$

Given that setup,

$$f(x|\lambda) = h(x)c(\lambda)\exp\left(w(\lambda)x\right)$$

The Casella/Berger format splits things into 3 factors,

1. A part that depends only on the data ($1/x!$)
2. A part that depends only on the parameter ($c(\lambda)$)
3. The interaction of the parameter and data in an exponential term ($\exp(w(\lambda)x)$.

$$p(x|\theta) = \underbrace{h(x)}_{\text{data only}} \quad \underbrace{c(\theta)}_{\text{parameter only}} \quad \underbrace{\exp(w(\theta)\,t(x))}_{\text{interaction (tilting)}}.$$

Example: Show that the binomial PMF belongs to the Binomial Exponential family.

$$f(x|p) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$= \binom{n}{x} (1-p)^n (1-p)^{-x} p^x$$

$$= \binom{n}{x} (1-p)^n \left(\frac{p}{1-p}\right)^x$$

$$= \binom{n}{x} (1-p)^n \exp\left(\log\left(\frac{p}{1-p}\right)x\right)$$

This is in the form of an exponential family, if we write the following,

$$h(x) = \begin{cases} \binom{n}{x} : x = 0, 1, 2, ..., n \\ 0 : \text{ otherwise} \end{cases}$$

$$c(p) = (1-p)^n$$

$$w_1(p) = \log\left(\frac{p}{1-p}\right), 0 < p < 1$$

$$t_1(x) = x$$

Identifying the pieces like that gives us,

$$f(x|p) = h(x)c(p)\exp[w_1(p)t_1(x)]$$

which is the form of an exponential family with $\theta = p$ and $k = 1$.

Example: Express the Normal distribution as a (two-parameter) exponential family.

Let $X \sim N(\mu, \sigma^2)$. The probability density function is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right].$$

Expand the exponent:

$$-\frac{(x-\mu)^2}{2\sigma^2} = -\frac{x^2}{2\sigma^2} + \frac{\mu}{\sigma^2}x - \frac{\mu^2}{2\sigma^2}.$$

Substituting,

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2} + \frac{\mu}{\sigma^2}x - \frac{\mu^2}{2\sigma^2}\right).$$

Rewrite by separating data-dependent and parameter-only terms:

$$f(x) = \frac{1}{\sqrt{2\pi}} \exp\left(\frac{\mu}{\sigma^2}x - \frac{x^2}{2\sigma^2}\right) \exp\left(-\frac{\mu^2}{2\sigma^2} - \frac{1}{2}\log(\sigma^2)\right).$$

Exponential-family form

The exponential family has the form

$$f(x \mid \eta) = h(x) \exp\bigl(\eta T(x) - A(\eta)\bigr).$$

Identify the components:

$$h(x) = \frac{1}{\sqrt{2\pi}}, \qquad T(x) = (x, x^2),$$
$$\eta_1 = \frac{\mu}{\sigma^2}, \qquad \eta_2 = -\frac{1}{2\sigma^2}.$$

The log-partition function is determined from the parameter-only terms:

$$A(\eta) = \frac{\mu^2}{2\sigma^2} + \frac{1}{2}\log(\sigma^2).$$

Expressing $A(\eta)$ in terms of η

Solve for μ and σ^2 in terms of η:

$$\sigma^2 = -\frac{1}{2\eta_2}, \qquad \mu = -\frac{\eta_1}{2\eta_2}.$$

Then

$$A(\eta) = -\frac{\eta_1^2}{4\eta_2} - \frac{1}{2}\log(-2\eta_2), \qquad \eta_2 < 0.$$

Final form

$$f(x \mid \eta) = \underbrace{\frac{1}{\sqrt{2\pi}}}_{h(x)} \exp\left(\underbrace{\eta_1 x + \eta_2 x^2}_{\eta^\top T(x)} - \underbrace{\left[-\frac{\eta_1^2}{4\eta_2} - \frac{1}{2}\log(-2\eta_2) \right]}_{A(\eta)} \right), \qquad \eta_2 < 0.$$

One reason exponential families are so important is that they naturally produce low-dimensional sufficient statistics. We will prove this rigorously later using the factorization theorem. For now, notice that whenever we can write a likelihood in exponential-family form, the data appear only through the functions $t_1(x), t_2(x), ..., t_k(x)$

Appendix: Proof of equivalence of the two forms

Theorem: Algebraic equivalence of the Wasserman and Casella/Berger forms.

Proof: We show that the two common ways of writing an exponential-family density/pmf are algebraically equivalent:

$$\textbf{(Wasserman / canonical form)} \qquad f(x \mid \eta) = h(x) \exp\{\eta \cdot T(x) - A(\eta)\}, \tag{1}$$

and

$$\textbf{(Casella–Berger / factorized form)} \qquad f(x \mid \theta) = h(x)\, c(\theta)\, \exp\{w(\theta) \cdot t(x)\}. \tag{2}$$

Here x is the observation, $(T(x), t(x))$ are vector-valued statistics, and $(\eta, w(\theta))$ are vector-valued parameters (the scalar case is the same argument).

(1) From Wasserman to Casella–Berger.

Assume (1) holds with a given mapping $\eta = \eta(\theta)$. Then

$$f(x \mid \theta) = h(x) \exp[\eta(\theta) \cdot T(x) - A(\eta(\theta))]$$

Define

$$w(\theta) := \eta(\theta), \qquad t(x) := T(x), \qquad c(\theta) := \exp\{-A(\eta(\theta))\}$$

.

With these definitions,

$$f(x \mid \theta) = h(x)\, c(\theta)\, \exp\{w(\theta) \cdot t(x)\},$$

which is exactly the Casella–Berger form (2). Thus any Wasserman representation yields a Casella–Berger representation by the simple identifications above.

(2) From Casella–Berger to Wasserman.

Assume (2) holds. Set

$$\eta(\theta) := w(\theta), \qquad T(x) := t(x).$$

Define a function A on the image $\mathcal{I} := \{\eta(\theta) : \theta \in \Theta\}$ by

$$A(\eta(\theta)) := -\log c(\theta), \qquad \theta \in \Theta,$$

(which is well-defined on $\mathcal{I}$ provided that whenever $\eta(\theta_1) = \eta(\theta_2)$ we have $c(\theta_1) = c(\theta_2)$; in practice one usually works with a reparameterization $\theta \mapsto \eta$ or restricts A to the image $\mathcal{I}$. Then

$$c(\theta) = \exp\{-A(\eta(\theta))\},$$

and substituting into (1) gives

$$f(x \mid \theta) = h(x) \exp\{\eta(\theta) \cdot T(x) - A(\eta(\theta))\},$$

which is the Wasserman canonical form (1) with $\eta = \eta(\theta)$ and A as defined.

Remark. The two forms are equivalent up to reparameterization: the Wasserman form emphasizes the **natural parameter** η and the log-partition function $A(\eta)$, while Casella–Berger emphasize an explicit multiplicative separation into $h(x)$, $c(\theta)$, and an exponential interaction. Algebraically the maps

$$w(\theta) \longleftrightarrow \eta, \qquad c(\theta) \longleftrightarrow e^{-A(\eta)}$$

provide the one-to-one dictionary between the representations (modulo the technical point about the domain/image of the mapping $\theta \mapsto \eta$).

Appendix - Exponential Tilting

We begin with a *baseline distribution* having density (or pmf)

$$f_0(x) = h(x) \exp\big(-A(0)\big).$$

This distribution assigns probability mass symmetrically according to its natural shape.

Applying an Exponential Weight

Let $T(x)$ be a statistic of interest. For a real parameter η, define a new density by multiplying the baseline density by an exponential weight:

$$\tilde{f}_\eta(x) = e^{\eta T(x)} f_0(x).$$

This operation *reweights* the distribution:

- If $\eta > 0$, larger values of $T(x)$ receive more weight.
- If $\eta < 0$, smaller values of $T(x)$ receive more weight.

Geometrically, this can be thought of as *tilting* the distribution so that probability mass shifts toward outcomes favored by $T(x)$.

Renormalization

The function $\tilde{f}_\eta(x)$ does not integrate to one. To obtain a valid density, we renormalize:

$$f_\eta(x) = \frac{e^{\eta T(x)} f_0(x)}{\mathbb{E}_0[e^{\eta T(X)}]} = h(x) \exp\big(\eta T(x) - A(\eta)\big),$$

where

$$A(\eta) = \log \mathbb{E}_0 \left[e^{\eta T(X)} \right].$$

The result is a new distribution in the *same family*, but with a different parameter value.

Interpretation

Exponential tilting does not distort the overall shape of the distribution. Instead, it systematically shifts probability mass:

- For continuous distributions (e.g. Normal), tilting often shifts the mean while preserving the variance.
- For discrete distributions (e.g. Poisson), tilting rescales the rate parameter.

Thus, an exponential family can be viewed as the collection of all distributions obtained by tilting a single baseline distribution in different directions.

Key Takeaway

Exponential tilting reweights a distribution so that outcomes with larger (or smaller) values of a statistic become more likely, while remaining within the same mathematical family.